

main.c



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Output

```
1 #include <stdio.h>
2 #define V 5
3
4 int minKey(int key[], int mstSet[]) {
5     int min = 9999, min_index;
6
7     for (int v = 0; v < V; v++)
8         if (mstSet[v] == 0 && key[v] < min)
9             min = key[v], min_index = v;
10
11     return min_index;
12 }
13
14 void printMST(int parent[], int graph[V][V]) {
15     printf("Edge \tWeight\n");
16     for (int i = 1; i < V; i++)
17         printf("%d - %d \t%d \n", parent[i], i, graph[i][parent[i]]);
18 }
19
20 void primMST(int graph[V][V]) {
21     int parent[V];
22     int key[V];
23     int mstSet[V];
24
25     for (int i = 0; i < V; i++) {
```

```
Edge    Weight
0 - 1    2
1 - 2    3
0 - 3    6
1 - 4    5
```

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```
26     key[i] = 9999;
27     mstSet[i] = 0;
28 }
29
30 key[0] = 0;
31 parent[0] = -1;
32
33- for (int count = 0; count < V - 1; count++) {
34     int u = minKey(key, mstSet);
35     mstSet[u] = 1;
36
37     for (int v = 0; v < V; v++)
38-         if (graph[u][v] && mstSet[v] == 0 && graph[u][v] <
            key[v]) {
39                 parent[v] = u;
40                 key[v] = graph[u][v];
41             }
42 }
43
44 printMST(parent, graph);
45 }
46
47- int main() {
48-     int graph[V][V] = {
49         {0, 2, 0, 6, 0},
50         {2, 0, 3, 8, 5}
```

Output

Clear

```
* Edge    Weight
0 - 1    2
1 - 2    3
0 - 3    6
1 - 4    5

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34

int u = minKey(key, mstSet);

35

mstSet[u] = 1;

36

37

for (int v = 0; v < V; v++)

38

if (graph[u][v] && mstSet[v] == 0 && graph[u][v] <

key[v]) {

39

parent[v] = u;

40

key[v] = graph[u][v];

41

}

42

}

43

44

printMST(parent, graph);

45

}

46

47

int main() {

48

int graph[V][V] = {

49

{0, 2, 0, 6, 0},

50

{2, 0, 3, 8, 5},

51

{0, 3, 0, 0, 7},

52

{6, 8, 0, 0, 9},

53

{0, 5, 7, 9, 0}

54

};

55

56

primMST(graph);

57

return 0;

58

}

Output

Edge

Weight

0 - 1

2

1 - 2

3

0 - 3

6

1 - 4

5

=== Code Execution Successful ===